

Predicting the Onset of Dementia Using Neural Networks

A Deep Learning Approach to Neuroimaging Analysis

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Abstract—Early identification of dementia is challenging due to heterogeneous clinical presentations and limitations in current diagnostic workflows, which are often reactive rather than predictive. This paper presents a modular deep learning framework for dementia risk prediction that integrates three complementary data modalities: structural magnetic resonance imaging (MRI), clinical biomarkers, and speech. For each modality, a dedicated neural model is developed and evaluated, allowing modality specific feature learning while maintaining flexibility for future research / integration into clinical applications.

The magnetic resonance imaging model leverages Convolutional Neural Networks to capture structural brain changes associated with disease progression. The biomarker model processes tabular clinical and vascular risk factors to provide a sensitive screening signal. In addition, a speech based multimodal pipeline is introduced that combines fine tuned acoustic and linguistic representations derived from pretrained speech and language models. These representations are fused using early concatenation and classified with a lightweight neural network.

Together, these findings support the effectiveness of modular and multimodal learning strategies for dementia prediction under limited data and compute constraints, while highlighting the need for broader validation on diverse clinical datasets.

I. INTRODUCTION

A. Background on Dementia

Dementia is a complex syndrome that poses a significant clinical public health challenge. As dementia remains without an effective cure, it is set to affect over 131 million people by the year 2050. [3] Dementia is defined as the acquired loss of cognition in multiple areas (memory, language, executive function) enough that it causes impairments with an individual's social or occupational performance. [3] Although age is a major risk factor, [5] for the symptoms to be properly defined as dementia, it must be a clear decline from a person's previous level of ability. [2]

One reason that dementia is of particular importance to us is because of its devastating impact on those that have it and those surrounding them as well. One way that we see this is dementia's ability to erode at core human needs, such as comfort, identity, inclusion and occupation [1]. The impact of dementia tends to go beyond that of the patient. It is often called the 'family disease' because of its impact on

family members. [8] Caregivers often report higher rates of depression, chronic fatigue and social isolation. [8]

This profound impact on the family is reflected in the economic burden family members are often required to shoulder. In 2019, a study found that over \$1.3 trillion dollars were spent worldwide on dementia. This can be split into 3 different categories: informal, social services and direct medical care. Informal care makes up 50% of the yearly cost and covers unpaid services like cooking, cleaning and other tasks caregivers do in service of the affected person. 34% is then in the form of social services, this includes things like nursing homes or care facilities. The other 16% is attributed to direct medical costs such as doctor visits or medication. [7]

A critical group that could benefit greatly from earlier intervention are stroke survivors. Experiencing a stroke doubles the likelihood of an individual later developing dementia. [4] The time it takes for stroke-survivors is often quick, as 31.8% of stroke patients developed dementia within 3 months of the stroke happening. [6] The reason for this is because of the ability of vascular damage to accelerate underlying Alzheimer's pathology. [4] Post-stroke dementia patients have an up to 6 times higher mortality rate than stroke survivors who do not develop dementia. [4]

Dementia does not have a known cure, and with our current technology we can only manage symptoms and slow its progression. [3] However, this does not mean that the situation is hopeless. If we are able to delay the onset of dementia by 5 years, we can reduce its prevalence by up to 4 million cases within 50 years. [9] The best way to do this is to diagnose dementia early. Nevertheless, there are plenty of challenges that we currently face when trying to do so.

B. Current Diagnostic Landscape

Currently, the diagnosis of dementia relies on a multi-step framework. The framework seeks to first confirm cognitive decline and then tries to identify the underlying etiology. The first step in the process is to collect patient history, with a reliable person to also corroborate what the patient is saying. Then, cognitive screening tools are given, like the MMSE to help identify dementia. Each patient also gets laboratory tests in order to rule out other illnesses that may be causing similar

symptoms. [2] One major part of these tests is neuroimaging. Guidelines have always recommended scans of the brain, through an MRI or CT, in order to rule out brain tumors or subdural hematomas, as they can present with similar symptoms. [13]

The current framework can only seek to confirm existing dementia or rule out other illnesses that have similar symptoms. The MMSE has a low diagnostic yield and often shows normal scores in patients with clear signs of dementia because it fails to look at executive function. [14] The MoCA screening tool was created to try to improve the diagnostic accuracy by taking into consideration other factors like execution functioning, which it successfully did. One study found that the AUC for MoCA was 0.934 vs MMSE's 0.772. [14] While it did improve the accuracy of the diagnostic framework, it is still severely limited in what it can do. It can only confirm present dementia. The current framework is reactive rather than proactive.

The current framework is not suited for dementia, as dementia is an incurable illness. The current treatments for dementia can only manage symptoms. [3] In order to get the most clinical benefit, we need a framework that can provide us with early intervention to slow its progression. There is a great clinical need for a predictive model rather than a better diagnostic test.

In order to predict future dementia, we can look at biological markers that may appear in patients before dementia develops. Since neuroimaging is already ordered for all patients to rule out other things, this is the perfect tool to use to predict the progression of dementia. Rather than just looking for tumors, an advanced model can look at the pathological signs like medial-lobe atrophy, white-matter changes and silent infarcts. [4]

Stroke survivors are an example of a high-risk patient population who could greatly benefit from a predictive model. A stroke doubles the risk of developing dementia and nearly a third do develop it within 3 months. [4] [6] Within this group, we can look at a variety of different data to determine if dementia would develop. Imaging data can provide the model with the stroke's location, severity and also show an underlying vascular disease. A model can also look at other clinical data, such as comorbidities like diabetes or hypertension. [9] Another tool that can be used to predict dementia is the presence of dysphasia. [6]

C. Thesis Statement & Research Question

This paper introduces a new deep learning framework that combines two different types of data, structural MRI scans of the brain and acoustic speech features, in order to predict the onset of dementia. While previous studies were successful at predicting dementia with just neuroimaging [24] or just speech analysis [22] alone, there is yet to be a study that uses both of these modalities in one model to make a prediction. We built a specific architecture that pulls MRI data from the ADNI dataset and speech audio from DementiaBank, designed to take advantage of the unique information revealed from these two modalities (structural changes vs functional decline)

and how they complement each other in one model to make a prediction. We believe this multimodal approach that uses both input modalities will beat unimodal baselines because it detects structural changes in the brain as well as early language impairments that appears in speech. [21] [23] Beyond just more accurate predictions, we aim to analyze which features matter most in predictions, seeking to create interpretable data that clinicians can actually use in the medical decision making process.

D. Overview of the Paper

The remainder of this paper is organized as follows. Section 2 provides a comprehensive literature review, beginning with early computational approaches such as Support Vector Machines (SVM) and transitioning into the current state-of-the-art deep learning methods for neuroimaging and speech analysis. Section 3 details our methodology, including the specific datasets used (ADNI for MRI, DementiaBank for speech), our preprocessing pipelines, and the architecture of our multimodal neural network. Section 4 presents the results of our model, evaluating its performance against unimodal baselines. Finally, Section 5 discusses the clinical implications of our findings, acknowledges limitations, and proposes directions for future research in multimodal dementia detection.

II. RELATED WORKS

A. Early Computational Approaches in Neurology

Before deep learning, researchers had to rely on "hand-crafted features" and traditional classifiers. [15] This meant that domain experts had to manually identify what features to measure (hippocampal volume, cortical thickness). The features would then be fed into classifiers like SVMs or Random Forests. When we manually extracted certain features from data, this is called feature engineering. This is different than feature learning, where the model can look through data and automatically identify features that correlate with certain outcomes. [16]

SVMs were the gold standard for medical image classification. They work by finding a hyperplane that is best at separating distinct classes. One study was able to achieve 96% accuracy when looking at whole-brain MRI scans. [24] These results were also confirmed by post-mortem pathology. This proved that computers can be used to make predictions with comparable accuracy to radiologists.

Random forests are an ensemble learning method that creates multiple decision trees while training and then looks at the data to see which tree classifies the data points the best. An advantage of this method is that it is more robust when given noisy data when compared to single decision trees. They are also better at handling non-linear relationships better than SVMs. One study used MMSE scores and was able to achieve an 86% accuracy with their model. [26] This served as a baseline to be compared to deep learning models.

One critical limitation with traditional machine learning techniques is that they require prior knowledge of what you want to look for. [15] Manually extracting features is time

consuming and is limited by expert knowledge. [18] These methods often struggle with high-dimensional data, like full MRI scans. [15] This means that if an expert does not know about a feature or is unable to quantify it, the model is unable to pick up on it and use it in its prediction. [16] These constraints shift the focus from traditional machine learning to deep learning.

B. Deep Learning Foundations for Medical Imaging

The limitations of the current diagnostic framework demonstrates a clear need for new tools to improve accuracy. Machine learning can be used to analyze medical images, but it requires "hand-crafted" features. Analyzing these medical images requires clinicians to know what to look for in order to feed it to a classifier. This process takes a lot of time and is limited by the expert's prior knowledge. [15] [18] Deep-learning introduces a different approach. Instead of manual feature extraction, the model can take the image and look for what feature needs to be extracted. [18] This allows features that experts might have missed or might not have been able to quantify to actually be taken into consideration and to be learned by the model. [15] [16]

Convolutional Neural Networks (CNNs) are currently the standard deep learning architecture for medical imaging. This is because of their ability to preserve spatial relationships in their learning. [17] CNNs are able to do this through the use of "convolutional layers" which act as filters. Early layers extract simpler features, like lines or edges. While deeper layers are able to combine these features in order to recognize more complex shapes like anatomical structures. [16] [17] CNNs are also more efficient when compared to other fully connected networks because of the use of "weight sharing" and "local receptive fields". Both of these significantly reduce the number of parameters, which makes it computationally more efficient to train when using them with very high resolution medical images. [15]

One issue with training these models is that they require very large datasets, and medical datasets tend to be very small. [15] In order to combat this, researchers use a technique called "transfer learning". This is a technique where neural networks are trained on general datasets to learn basic visual features. They are then fine-tuned based on the smaller medical dataset in order to be able to learn more specific features. This has proven successful and models are very effective at detecting lymph nodes and classifying lung diseases. [17] [19] [15] There is also a technique called data augmentation where images can be rotated, flipped or scaled in order to artificially increase the dataset size and avoid overfitting. [16]

Transfer learning is particularly valuable for Alzheimer's detection because MRI datasets are often very small (hundreds of scans) when compared to other computer vision datasets which contain millions of images. [28] [29] This method works based on the fact that the lower layers of CNNs capture generic features (like edges and shapes) that can be applied to many types of images. With this in mind, only the top layers need to be retrained for medical specific features. [28]

What this learning approach does is that it opens up the ability for researchers to use pre-trained architectures like VGG16 or Inception that already gained "knowledge" from their training on ImageNet. [28] Research shows that this method is not only able to achieve comparable, and sometimes better, performance than training from scratch, but it also requires far less data and computational resources. [28] [29] One study found that even with training data 10 times smaller than state-of-the-art methods, models based on VGG16/Inception architecture that used transfer learning techniques were able to achieve comparable results using the OASIS MRI dataset. [28] The ability to successfully transfer knowledge gained from training on natural images to medical images proves that this technique is valuable for dementia prediction models that are working with limited datasets. [28] [19]

Deep learning models have achieved exceptional performance, outperforming traditional methods, in segmenting complex structures, such as brain tumors or lesions from multiple sclerosis. [15] [17] CNNs have already demonstrated they are able to detect skin cancer at a similar accuracy when compared to dermatologists. They have also greatly reduced the amount of false positives when detecting lung nodules compared to CAD systems. [16] [18] In brain imaging specifically, deep learning techniques have proven successful in classifying Alzheimer's vs Mild Cognitive Impairment (MCI) by fusing MRI and PET data. [15] [20] [16]

C. Key Studies on Neural Networks and Dementia Prediction

Given the success of using CNNs in general medical imaging, researchers chose to apply them for predicting dementia. Most studies use structural MRI from datasets like ADNI. Studies vary and can usually be categorized by: the data modality that was used, the architecture design and the classification task.

Early work has proved that automated classification could achieve similar accuracy when compared to radiologists. [24] This model used SVM on whole-brain MRI images. Their results were also validated with post-mortem pathology. [24] The main limitation of this model was that it required manual extraction of features. Later work was able to use deep learning to combine MRI and PET data. [20] This multimodal approach was able to achieve better performance compared to the single-modality approach. This showed that multimodal models can be used to improve accuracy, but this specific study only combined imaging modalities and did not explore other types of data.

At the same time as the neuroimaging research, other researchers were exploring the potential for speech to be used as a biomarker for dementia. One study found that language impairments appear months before memory symptoms. [21] In this study, narrative speech samples were analyzed and researchers were able to identify distinct linguistic features that were able to distinguish AD from healthy controls. [21] Another meta-analysis of speech features found that "speech rate" and "pauses" both have proven effect in detecting AD. [23] This provides us with evidence-based biomarkers to

use for our model. Another study was able to apply deep learning to the dementiabank dataset. [22] They were able to achieve high accuracy using speech and transcripts. This demonstrated the potential for automated speech analysis, but these analyses were done in isolation and could greatly benefit when combined with neuroimaging.

Literature shows us two different ways to identify dementia, neuroimaging and speech/cognitive studies, both of which were able to achieve high accuracy. However, there are few studies that were able to combine both imaging and non-imaging modalities into one model. This is a problem because each modality captures different aspects of pathology. MRI data shows structural brain changes while speech shows functional cognitive decline. A model that combines these two would be able to allow us to predict dementia earlier.

D. Multimodal fusion techniques

In order for physicians to make accurate medical diagnoses, physicians must look at various kinds of data (imaging, lab results, notes, vital signs). Multimodal learning is when different types of data are combined in order to improve prediction. However, each modality has different representation, scale and dimensionality. Images, for example, contain high-dimensional spatial data. Speech, on the other hand, has temporal sequential data. Clinical scores contain low-dimensional numerical data. Each type of data requires different types of analysis to extract features from them. Fusion of these data types seeks to capture complementary information from all of these datasets that a unimodal model would miss.

There are two different approaches to fusing data. The first type is called early fusion, which is also known as "feature-level fusion". The idea with this approach is to combine raw data, or low-level features, from different modalities before you feed it to the model. The main advantage with this approach is that the model learns joint representation from the start. This approach is difficult because it requires precise alignment and normalization between the two modalities. An example of this is mixing 3-dimensional MRI voxels with clinical scores. The second approach is late fusion, also known as "decision-level fusion". The idea with this approach is to train separate models for each modality, and then combine their predictions afterward. The advantage with this approach is that each modality is processed with specialized architecture. So, CNNs are used for images while RNNs are used for sequences. This is much more flexible than the early fusion approach and can handle missing modalities. So you can train the CNN on MRI data and RNN of speech data separately, and then simply combine their output probabilities separately.

There is a third approach called intermediate fusion, also called "hybrid fusion". This approach extracts features from each modality separately in early layers. The feature representations are then combined in middle layers. The process continues with shared layers for the final prediction. This approach balances the benefits of both early and late fusion. One study was able to successfully use this approach by fusing MRI and PET data. [20] The more modern technique is called

attention-based fusion. With this approach, the model learns to weight different modalities dynamically. Some modalities may be more informative for certain patients, and the model is able to pick up on this. The advantage of this technique is that it is interpretable, and we can see which modality drove each prediction. This is also valuable because of its ability to establish clinical trust and can lead to more widespread adoption of the model.

In dementia research, most multimodal work combines two different imaging modalities (MRI+PET or MRI+fMRI). An example of this is a study that fused MRI and PET using hierarchical deep learning. [20] This model was able to achieve better performance than a model with a single modality. This model used the intermediate fusion approach. This model was also limited to imaging data only. There are few studies that combine imaging data with non-imaging data. Specifically, the fusion of MRI and speech has not been explored yet for dementia. This gap in research motivates our fusion approach in this work.

E. Speech and Language Analysis in Dementia Detection

Early research focused on extracting specific linguistics markers from narrative speech to identify cognitive decline. Fraser et al. analyzed speech samples collected from 240 AD patients and 233 healthy controls in order to investigate this approach. Through their analysis, 4 distinct factors of impairment were identified: semantic impairment, acoustic abnormality, syntactic impairment and information impairment. [21] This study established that linguistic analysis of patients can be used to reliably classify them into healthy and AD group [21]

A recent systematic review of 11 studies specific acoustic features as a reliable way to classify dementia. [23] The meta-analysis categorized these features into three groups: temporal, rate and interruption. The study found that rate and interruption related parameters were the most significant in differentiating between groups. ($p \leq 0.01$) [23] Specifically, healthy controls were found to speak at significantly faster rates, with a mean difference of 0.64 ($p = 0.0100$) and articulation rate ($p = 0.0002$) [23] They also found that AD patients showed more vocal instability, or voice breaks, with a mean difference of -11.58% ($p < 0.0001$) [23]

Newer approaches were able to apply deep learning techniques to the DementiaBank dataset to deal with multimodal data. Ilias et al. proposed new architectures that combined BERT for text processing with Vision Transformers for log-Mel spectrograms. [22] They were able to introduce a new mechanism, "Multimodal Shifting Gate", which dynamically weights the importance of different modalities while it is processing. [22] Through this approach, they were able to achieve state-of-the-art performance, with a classification accuracy of 90.00% and 3.61 RMSE in predicting MMSE score. [22]

Despite these successes, existing speech models still have limitations. Most studies train separate models for acoustic and linguistic features and rely on late fusion fusion strategies like majority voting. These simple fusion strategies fail to capture the complex, non-linear, patterns between the two features.

[22] More importantly, speech studies are mostly done in isolation from neuroimaging studies. While speech analysis detects functional cognitive decline [21] and MRI detects structural pathology [4] there is no framework yet that combines these two modalities to make a prediction. This gap represents an opportunity to leverage the complementary nature of both modalities for earlier and more accurate predictions.

F. Interpretability and Explainability in Medical AI

As these deep learning models become more sophisticated and accurate, even though they achieve increasingly high accuracy, another problem that arises is that these models are often black boxes. For a physician to be able to accept a prediction and use it in healthcare, a physician must understand how a model was able to come to that prediction. [11] For these complex deep learning models to be trusted, their methods must be interpretable. This section will explore the importance of explainability in medical AI and methods for achieving interpretable results.

There are many reasons why interpretability in AI is important. One of them is medical data being incredibly complex and often incomplete. Medical decision making involves a lot of information and uncertainty and a clinician needs to understand the reasoning behind the prediction especially if it goes against their intuition. [11] The second reason is that regulatory requirements and ethics require transparency in medical AI systems. [11] The third reason is that for widespread adoption of AI in clinical practice, interpretability can lead to increased trust. [11] But, this brings about a conflict, as oftentimes the most accurate models (like neural networks) are the least transparent and the most transparent models (like decision trees) are the least accurate. [11] This brings into question whether the trade-off between accuracy and transparency is worth it and where is the best place to draw the line.

A distinction has to be made between explainability and causability. [11] Explainability refers to methods which highlight specific, decision-relevant, aspects of the algorithm to see which parts actually contribute to the prediction. [11] This is a property of the machine itself. Causability, however, is a property of a person. It refers to the extent to which an explanation can help a human expert gain an understanding of the cause. [11]

This difference is important because ML models use correlation/probability to make predictions while doctors use reasoning, understanding and often seek causal relationships. [11] For an AI to be used clinically, it can not just be technically accurate, it must be able to offer causal relationships in a way that is meaningful to clinicians. [11] This requires that models either incorporate causal structure into its design or a way to map technical explanations to causal models that can be understood by humans. [11]

The methods for achieving interpretability can broadly be categorized into two categories: ante-hoc (built into the model) vs post-hoc (applied after training). [11] Ante-hoc (glass box) models are interpretable by design, such as linear regression,

decision trees or fuzzy inference systems. [11] These designs are inherently transparent and any expert can look at it and understand what it is doing. But, to get this transparency, often accuracy is sacrificed. Post-hoc (black box analysis) methods are applied to already trained models, such as black-box models like neural networks. [11] Some common post-hoc techniques include gradient based methods, which analyzes how inputs affect the output, and attention mechanisms, which focuses on which parts of the input the model gives the most attention to when it is making predictions. [11]

For medical imaging, saliency maps and attention visualizations have proven to be valuable. These methods can be used to highlight which brain regions a model uses when it is classifying dementia, which allows clinicians to verify that the model is focusing on areas of the brain that are clinically relevant. [11] In multimodal models, attention mechanisms can help determine which input modality contributes more to each prediction, which provides insight into the relative importance of each data type. [22] This is very relevant to our multimodal model as we want to investigate how structural changes in the brain and functional decline drive predictions and see how they could inform medical decision making.

The interpretability of our model is important for several reasons. The first of them is that by identifying specific features that contribute the most to predictions, we can ensure that the model is using clinically relevant biomarkers. Second, if the model is interpretable, we can apply what we learned in order to identify patients that might benefit more from earlier intervention, as we can see which combination of risk factors the model considers significant. Third, for high-risk groups like stroke survivors, understanding which features predict dementia can help guide targeted screening as well as monitoring strategies. [4]. Lastly, a model being interpretable is important in building clinical trust, which is essential for translating these research models into clinical practice.

G. Identifying the Research Gap

Traditional machine learning achieved good accuracy, around 86-96%, but it requires manual feature extraction. [24] [26] CNNs were able to improve automation and achieved success in medical imaging tasks. [15] [16] Multimodal imaging analysis was able to outperform other unimodal approaches. [20] Speech biomarkers have also been used and demonstrated their value in making predictions. [21] [23] [22] Despite all of these advances in prediction, a model that integrates all of these advances still has not been done.

There is no existing work that combines both structural data from neuroimaging with acoustic speech biomarkers for predicting dementia. MRI is able to capture structural pathology like atrophy, white matter changes and vascular damage. [4] Speech is able to show functional impairment prior to a clinical diagnosis. [21] In order to make an early prediction, a model that detects both structural precursors as well as functional changes is required. Both of these modalities provide complementary information. MRI shows anatomical substrate of disease while speech shows the real-world cogni-

tive impact. A fusion of the two would be better at showing a fuller picture of the disease state. High risk populations like stroke survivors would benefit from this multimodal prediction approach. [4] [6]

This paper will address this gap by developing a multimodal deep learning framework. This framework will be the first to fuse structural MRI data with acoustic speech features for dementia prediction. This model will use proven datasets, ADNI and DementiaBank. This paper will also investigate the importance of different features to identify specific biomarkers that drive predictions. This model will provide a pathway towards earlier intervention, particularly for high-risk populations.

III. METHODOLOGY

A. System Overview

This study employs a multimodal approach consisting of three independent predictive models:

- (1) an MRI-based model,
- (2) a biomarker-based model,
- (3) a speech-based model.

Each model processes a different modality of patient data and outputs a predicted dementia severity stage. The following subsections describe the full methodology for each model, including data preprocessing, model design, training strategy, and evaluation metrics.

B. MRI Model

1) *Dataset and Preprocessing*: The MRI dataset consisted of over 68.5k+ structural brain imaging samples, each associated with a corresponding dementia stage category [27]. Prior to model development, the dataset utilized various preprocessing techniques to ensure consistency, variety, enhanced training, and model readiness.

In order to add additional variation into the dataset, all images within the dataset were first converted to grayscale and resized to a standardized 128×128 resolution. To improve model generalization, data augmentation was applied during training, including random horizontal flipping and small rotations, while test images were left unaugmented. The dataset was split into training and testing subsets at an 80:20 ratio.

2) *Configuration*: The MRI Model was initialized to prioritize training / testing only when GPU acceleration is available, and data loading and preprocessing leverages multiple available CPU cores in parallel to further optimize training efficiency. Training was performed using a batch size of 64 for 5 epochs with a learning rate of 1×10^{-4} , providing a balance between optimization stability and convergence efficiency.

3) *Model Architecture*: The MRI classifier is implemented as a custom lightweight Convolutional Neural Network (CNN) optimized specifically for 128×128 structural MRI slices. The architecture begins with an initial Batch Normalization layer to stabilize voxel-intensity variation across MRI scans. This is followed by two convolutional stages: the first applies a 3×3 convolution with 16 feature maps, while the second increases the depth to 32 feature maps. Each convolution is paired

with a ReLU activation and a 2×2 max-pooling operation, reducing spatial resolution from 128×128 to 32×32 while progressively capturing higher-level structural features such as cortical boundaries and texture differences linked to dementia progression.

The resulting feature maps are flattened into a single 32-32-32 feature vector and passed through a fully connected classification head. The first dense layer generates a 128-dimensional intermediate representation and incorporates a 0.4 dropout rate to mitigate overfitting. A final linear layer produces the four output logits corresponding to the dementia classes: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia.

4) *Training Procedure*: Model training followed a supervised learning workflow in which the network parameters were optimized to minimize categorical prediction error. Each training epoch consisted of iterating through mini-batches of MRI slices, performing a forward pass through the CNN to generate class logits. These outputs were evaluated using the cross-entropy loss function, which quantifies the divergence between predicted probabilities and true dementia labels.

Gradient-based optimization was carried out using the Adam optimizer with the initial learning rate of 1×10^{-4} . For each batch, gradients were computed via backpropagation, and Adam updated the weights using its adaptive moment estimates to promote stable and efficient convergence. Dropout regularization remained active within the fully connected layer during training, reducing co-adaptation among neurons and mitigating overfitting.

5) *Handling Class Imbalance*: The MRI dataset exhibited imbalance across dementia severity stages, with some classes appearing significantly more frequently than others. To correct for this, two strategies were implemented. First, class weights inversely proportional to class frequency were applied in the loss function, ensuring that errors on minority classes contributed more strongly during gradient updates. Second, a WeightedRandomSampler was used to oversample minority-class examples during training, creating more balanced batches and reducing the tendency for the model to favor dominant classes. Together, these techniques improved overall classification performance and prevented the model from collapsing toward majority-class predictions.

6) *Evaluation Metrics*: During training, computational performance was monitored by measuring the total time required to complete all five epochs. Overall model performance was assessed using standard multi-class evaluation metrics, including accuracy, precision, recall, and F1-score. Accuracy provided a general measure of overall correctness, while precision and recall offered insight into the model's performance across individual dementia stages. The F1-score allowed for a more balanced evaluation in the presence of class imbalance. These metrics were computed on the held-out test set to provide an unbiased estimate of real-world model performance.

C. Biomarker Model

1) *Dataset and Preprocessing*: The biomarker dataset consisted of 1.8k+ patient-level clinical measurements, including demographics, vascular risk factors (e.g., diabetes, hypertension, hypercholesterolemia), MRI-derived biomarkers (e.g., lacunes, microbleeds, Fazekas score), and cognitive indicators. The target variable was a binary dementia label.

All preprocessing was performed programmatically to ensure consistent numerical formatting across heterogeneous clinical fields. Text-based categorical variables such as gender and smoking status were normalized to integer encodings, while yes/no-type attributes were converted to binary indicators using an extensive mapping to capture common formatting differences (e.g., “Present,” “Y,” “NO,” etc.). Non-numeric values were coerced to numeric form and missing entries were imputed with zeros.

After cleaning, the dementia label was isolated as the prediction target, and all remaining clinical features were scaled to the [0,1] range using MinMax normalization. The dataset was split into training and testing subsets with an 80:20 ratio to ensure reliable model evaluation.

2) *Configuration*: Model training was executed with GPU acceleration when available, while CPU fallback ensured compatibility on all systems. The dataset was wrapped in PyTorch Dataset and DataLoader objects for efficient mini-batch training. A batch size of 8 provided stable gradient updates for the tabular input. The training process spanned 25 epochs using the Adam optimizer with a learning rate of 1×10^{-3} . To standardize output performance measurement, the total training time was recorded from the start of epoch 1 to the end of epoch 25.

3) *Model Architecture*: The biomarker classifier is implemented as a lightweight Convolutional Neural Network tailored for tabular clinical data. The model receives a flattened feature vector of patient biomarkers and processes it through a three-layer feedforward architecture.

The first hidden layer expands the input into a 32-dimensional representation, followed by ReLU activation and a 0.15 dropout rate. A second hidden layer reduces the representation to 16 units, again followed by ReLU activation and dropout. A final linear layer maps to a single output neuron, followed by a sigmoid activation that produces a probability of dementia. This compact architecture ensures efficient learning while avoiding unnecessary complexity for structured clinical inputs.

4) *Training Procedure*: Training was performed using supervised learning with binary cross-entropy loss, appropriate for the two-class dementia prediction task. For each batch, model weights were updated using Adam optimization, and the average loss per epoch was tracked to monitor convergence behavior. All computations were carried out on GPU hardware when available. To quantify computational performance, the full end-to-end training time for all 25 epochs was captured and reported.

5) *Evaluation Metrics*: Model evaluation was performed on the held-out test set using standard metrics for binary

classification, with the ‘Dementia’ label removed during training. Accuracy was computed to measure overall prediction correctness, while a full classification report—including precision, recall, and F1-score for both dementia and non-dementia classes—provided deeper insight into the model’s clinical usefulness. Predictions were produced by thresholding the model’s sigmoid output at 0.5, and results were printed for transparent comparison across classes.

D. Speech-Based Model

This section describes the complete methodology for the speech-based dementia detection pipeline. The approach is multimodal: we fine-tune two modality-specific encoders (wav2vec2 for audio and BERT for text), extract per-recording embeddings from each encoder, fuse the embeddings by concatenation, and train a compact multilayer perceptron (MLP) classifier on the fused representations. Implementation details, hyperparameters, and reproducibility notes are provided so the procedure can be replicated.

This work follows a single, end-to-end experimental flow: raw audio recordings and their corresponding transcripts are first preprocessed (audio resampled to 16 kHz and transcripts cleaned with a deterministic rule-based script). Two pretrained encoders—wav2vec2 for acoustic modeling and BERT for linguistic modeling—are then fine-tuned separately on the binary dementia/control labels to adapt their representations to the Cookie Theft domain. After fine-tuning, each encoder is used as a feature extractor to produce a fixed-length embedding for every recording (768-d from each encoder via mean- or pooler-based pooling); the audio and text embeddings are concatenated to form a 1536-d multimodal representation for each sample. These fused embeddings are persisted to disk and used to train a lightweight MLP classifier (80/20 train/test split, seed=42) whose weights are selected by highest observed test accuracy. All intermediate artifacts (fine-tuned checkpoints, processors/tokenizers, fused embeddings, and best classifier weights) are saved to enable reproducible inference and downstream analysis.

1) Dataset and Preprocessing:

a) *Dataset*: All experiments use the DementiaBank *Cookie Theft* subset. After transcript cleaning and matching audio-transcript pairs, the final dataset contains **551** paired samples (each sample = one audio recording + its cleaned transcript). Samples are labeled as *Dementia* or *Control*.

b) *Transcript cleaning*: Transcript cleaning was performed with a deterministic, rule-based script. The script:

- Extracts participant utterances (lines beginning with the participant marker).
- Removes CHAT-specific markup and ancillary tokens (e.g., angle-bracket annotations <...>, square-bracket comments [...]) and parenthetical comments (...).
- Removes disfluency markers of the form &... and other special markers (+...).
- Strips timestamps and standalone numbers, retains common punctuation (.,?!') and collapses repeated whitespace.

- Discards empty transcripts.

This cleaned text is used as the input to the BERT tokenizer / model.

c) Audio preprocessing: Audio files are resampled (or loaded) at $sr = 16,000$ Hz using `librosa`. No additional filtering or voice activity detection is applied beyond the default preprocessing performed by the `wav2vec2` processor during batching/padding.

2) *Overview of the Multimodal Pipeline:* The pipeline has four stages:

- 1) Fine-tune `wav2vec2` on the audio labels (binary classification).
- 2) Fine-tune BERT on cleaned transcripts (binary classification).
- 3) Extract utterance-level embeddings from the fine-tuned encoders and concatenate audio + text embeddings to form a fused representation (1536-d).
- 4) Train a lightweight MLP classifier on the fused embeddings (80/20 train/test split) and select the best model by highest test accuracy.

3) *Encoder Fine-Tuning:* Two encoder models were fine-tuned separately on the Cookie Theft dataset: a `wav2vec2` audio encoder and a BERT text encoder. Both encoders are fine-tuned end-to-end (all weights trainable) to adapt pretrained representations to the dementia classification objective.

a) Wav2vec2 fine-tuning (audio):

- Base model: `Wav2Vec2ForSequenceClassification` initialized from a local checkpoint of a pretrained `wav2vec2` model.
- Input: raw waveforms at 16 kHz; the processor pads and formats inputs for the model.
- Labels: binary (1 = Dementia, 0 = Control).
- Training configuration:
 - Logical batch size: 8
 - Micro-batch size (GPU memory fit): 1 (micro-batching with gradient accumulation)
 - Learning rate: 2×10^{-5}
 - Optimizer: AdamW
 - Scheduler: linear warmup followed by linear decay; warmup steps = 6% of total training steps
 - Number of epochs: 5
 - Validation split: 15% (random split)
 - Mixed precision: automatic mixed precision (AMP) with `torch.cuda.amp (autocast + GradScaler)`
 - Memory optimization: gradient checkpointing enabled when supported
 - Random seed: 42 (controls deterministic splitting and sampling where applicable)
- Loss & metric: Cross-entropy loss during training; validation accuracy is used to select the best checkpoint.
- Notes: Micro-batching and gradient accumulation are used to emulate a larger batch and fit computation on limited GPU memory. AMP and gradient checkpointing

reduce memory footprint and are noted for reproducibility.

b) BERT fine-tuning (text):

- Base model: `BertForSequenceClassification` initialized from a local pretrained BERT checkpoint.
- Input: cleaned transcripts tokenized with BERT tokenizer; sequences truncated/padded to `max_length=512`.
- Training configuration:
 - Batch size: 16
 - Learning rate: 2×10^{-5}
 - Optimizer: AdamW
 - Scheduler: linear warmup then linear decay; warmup = 6% of training steps
 - Number of epochs: 5
 - Validation split: 15%
 - Random seed: 42
- Loss & metric: Cross-entropy loss; validation accuracy used for checkpoint selection.

4) *Embedding Extraction and Fusion:*

a) Encoder outputs & pooling choices.: After fine-tuning, the encoders are used as feature extractors (feature mode) to produce fixed-length vectors for each recording:

- **wav2vec2 (audio):** Extraction uses the base `Wav2Vec2Model` (weights loaded from the finetuned directory). From the model output, the sequence-level representation is obtained by mean pooling the last hidden states across time: $\mathbf{v}_{\text{audio}} = \text{mean}(\text{last_hidden_state}, \text{dim} = 1)$. This produces a 768-dimensional audio embedding per recording.
- **BERT (text):** Extraction uses `BertModel` (weights from the finetuned directory). If `pooler_output` is present, it is used as the text embedding; otherwise the last hidden state is mean-pooled across tokens. This yields a 768-dimensional text embedding per transcript.

b) Fusion.: The two 768-d vectors are concatenated to form a $\mathbf{v}_{\text{fused}} \in \mathbb{R}^{1536}$:

$$\mathbf{v}_{\text{fused}} = [\mathbf{v}_{\text{text}} \parallel \mathbf{v}_{\text{audio}}]$$

All fused embeddings are saved to disk (file: `combined_new_embedding.pt`) for downstream classifier training. Samples with missing or empty transcripts are omitted during this stage, which results in the final dataset size of 551 embeddings.

5) *Classifier Architecture (MLP):* A compact MLP is trained on fused embeddings to perform the final binary classification.

a) Architecture.: The MLP is intentionally lightweight to reduce overfitting on the modest dataset size:

- Input dimension: 1536
- Dense layer 1: $1536 \rightarrow 256$
 - Layer normalization
 - ReLU activation
 - Dropout(p=0.35)

- Dense layer 2: $256 \rightarrow 64$
 - Layer normalization
 - ReLU activation
 - Dropout($p=0.25$)
- Output layer: $64 \rightarrow 2$ (logits for binary cross-entropy with logits via `CrossEntropyLoss`)

6) Training Procedure for the Classifier:

a) Data split and batching.:

- The fused embedding set is converted to a `TensorDataset` and shuffled.
- A random 80/20 train/test split (controlled by `seed = 42`) is used. The training set is used to update model parameters and the held-out test set is used to evaluate and select the best model each epoch.
- Batch size: 32

b) Optimization and schedule:

- Loss: `CrossEntropyLoss`.
- Optimizer: Adam (PyTorch default) with learning rate 3×10^{-4} and weight decay 1×10^{-6} .
- Number of epochs: up to 90 epochs. The training loop monitors test accuracy every epoch and saves the model state when test accuracy improves (best model checkpointing).
- Early stopping: implicit via best-model checkpointing (the code retains the state with the highest observed test accuracy). No formal patience-based early-stopping rule is used.

7) *Handling Class Imbalance*: No explicit imbalance correction was applied:

- No class weights in the loss function.
- No oversampling or synthetic sample generation.

Class imbalance (if present) is therefore a potential source of bias; this choice is deliberate and will be discussed alongside results and limitations.

8) Evaluation Metrics:

- The primary evaluation metric used in both encoder validation and classifier selection is **accuracy** (computed with `sklearn.metrics.accuracy_score`).
- Precision, recall, and F1-score were not computed in the current pipeline but can be added easily from predictions on the test set if desired.
- During encoder fine-tuning, validation accuracy is used to save the best encoder checkpoints. During MLP training, test accuracy is used to select the best classifier checkpoint.

9) Implementation Details and Reproducibility:

a) *Environment and libraries*.: Experiments were executed in Google Colab with GPU acceleration. Key libraries and tools:

- PyTorch (modeling/training)
- `transformers` (Hugging Face; Wav2Vec2, BERT, tokenizers, processors)
- `librosa` (audio I/O and resampling)
- `scikit-learn` (metrics)
- `numpy`, standard Python I/O for dataset handling

b) *Memory- and compute-aware training*.: To make fine-tuning feasible within GPU memory constraints, the wav2vec2 fine-tuning script employs:

- **Micro-batching + gradient accumulation**: logical batch size 8 implemented via micro-batches of size 1 and manual accumulation.
- **Automatic mixed precision (AMP)** using `torch.cuda.amp` (autocast + GradScaler) to accelerate training and reduce memory usage.
- **Gradient checkpointing** when supported by the model to trade compute for memory.

These measures are explicitly noted in the methodology because they affect training reproducibility and effective batch sizes.

c) *Random seeds and deterministic behavior*.: The same random seed (`seed=42`) is used across data splitting and model initialization to stabilize training runs. Exact reproducibility may still vary across systems and CUDA/cuDNN nondeterminism; nevertheless, the seed ensures consistent splits and initial conditions on a single environment.

d) *Saved artifacts*.: All key artifacts are saved to Google Drive during experiments:

- Fine-tuned wav2vec2 checkpoint and processor (directory: `finetuned_wav2vec`).
- Fine-tuned BERT checkpoint and tokenizer (directory: `finetuned_bert`).
- Fused embeddings file: `combined_new_embedding.pt`.
- Best MLP classifier weights: `best_mlp_classifier.pt`.

These checkpoints are preserved to allow downstream inference and to reproduce the embedding extraction and classification steps.

10) *Hyperparameter Summary*: The most important hyperparameters used in the pipeline are summarized in Table I.

11) *Reproducible Workflow (Ordered Steps)*: To reproduce the experiments exactly:

- 1) Clean transcripts using the provided cleaning script (retain only participant lines; remove CHAT markup).
- 2) Arrange audio and transcript files into per-class folders and verify matching base filenames.
- 3) Fine-tune wav2vec2 on audio labels (script: `finetune_wav2vec_colab_safe.py`); save best checkpoint.
- 4) Fine-tune BERT on cleaned transcripts (script: `finetune_bert_colab.py`); save best checkpoint.
- 5) Load finetuned encoders and extract embeddings per recording; mean-pool encoder outputs to produce 768-d vectors and concatenate to 1536-d fused vectors; save fused embeddings (`combined_new_embedding.pt`).
- 6) Train the MLP classifier on fused embeddings using an 80/20 train/test split and monitor test accuracy each epoch; save best classifier weights.

TABLE I
KEY HYPERPARAMETERS FOR ENCODER FINE-TUNING AND CLASSIFIER TRAINING.

Component	Hyperparameter / Value
Dataset final size	551 samples
Random seed	42
Audio sampling rate	16,000 Hz
wav2vec2 fine-tune	Epochs = 5; logical batch size = 8; micro-batch size = 1; learning rate = 2×10^{-5} ; optimizer = AdamW; validation split = 0.15; AMP with GradScaler; gradient checkpointing enabled
BERT fine-tune	Epochs = 5; batch size = 16; learning rate = 2×10^{-5} ; optimizer = AdamW; validation split = 0.15; max sequence length = 512
Embedding pooling	wav2vec2: mean pooling of last hidden states (768-d); BERT: pooler output if available, otherwise mean pooling (768-d)
Fusion	Concatenation of text and audio embeddings to form a 1536-dimensional vector
MLP classifier	Input dimension = 1536; hidden layers = [256, 64]; ReLU activations; LayerNorm; dropout rates = [0.35, 0.25]; optimizer = Adam; learning rate = 3×10^{-4} ; weight decay = 1×10^{-6} ; epochs = 90; batch size = 32
Evaluation	Primary metric = accuracy (scikit-learn)

7) For unseen inference, repeat embedding extraction for new audio/transcript pairs and pass fused embeddings to the saved MLP model.

12) Notes on Design Decisions:

- **Why fine-tune both encoders?** Fine-tuning adapts pre-trained speech and language representations to domain-specific cues found in the Cookie Theft task (e.g., fluency, semantic drift, lexical choice) and has been shown to improve downstream classification performance compared with using frozen encoders.
- **Fusion strategy:** Early fusion by concatenation is simple and empirically effective for small-to-moderate datasets. It avoids adding many trainable parameters (as would a large multimodal transformer) and simplifies interpretability of downstream classifier design.
- **Compact MLP:** The MLP is intentionally small to limit overfitting on a dataset of 551 samples. Layer normalization and dropout are included to increase robustness.
- **Metric choice:** Accuracy was used as the primary metric to remain consistent across encoder and classifier selection steps. Additional metrics (precision/recall/F1) can be reported in the Results section as needed.

E. Limitations

We document several limitations observed during experiments, many of which are typical for research using DementiaBank Cookie Theft recordings:

- 1) **Dataset size and diversity:** The dataset is relatively small (551 paired samples) and narrowly focused on the Cookie Theft picture description task. The limited variety of acoustic environments and speaking styles reduces the generality of learned cues.
- 2) **Cross-dataset incompatibility:** An attempt to enlarge the training data by incorporating an additional dataset

(DEMO@care) failed owing to a language mismatch (Greek), which produced noisy cross-lingual embeddings and poor performance. Cross-lingual fusion requires careful preprocessing and language-aware modeling that was beyond scope.

- 3) **Data augmentation limitations:** Aggressive audio augmentation (to expand the dataset) produced degraded classifier performance and greater overfitting in our experiments. We attempted to synthesize $\approx 3,000$ augmented samples, but observed decreased accuracy and therefore did not adopt heavy augmentation in the final pipeline.
- 4) **Compute constraints and training shortcuts:** GPU memory limitations in Google Colab required trade-offs (micro-batching, mixed precision, gradient checkpointing). These measures allowed training but forced shorter fine-tuning runs and more conservative hyperparameter choices.
- 5) **Known characteristics of DementiaBank:** Prior literature and our own experiments show that audio-only fine-tuning often underperforms, that recordings are short and lack acoustic variety, and that wav2vec2 can overfit quickly on this dataset. These characteristics explain the observed training instability and limit external validity.

IV. RESULTS & OBSERVATIONS

TABLE II
CLASS-WISE PERFORMANCE METRICS FOR THE MRI MODEL

Dementia Class	Precision	Recall	F1-score	Support
Non-Demented	0.98	0.94	0.96	13453
Very Mild Dementia	0.79	0.90	0.84	2756
Mild Dementia	0.82	0.97	0.89	987
Moderate Dementia	0.97	0.99	0.98	92
Weighted Average	0.90	0.90	0.90	17288

1) *MRI Model:* Derived from the final evaluation, the MRI model achieved a testing accuracy of 94.97% after 5 training epochs. Performance analysis using a multi-class classification report yielded average precision, recall, and F1-score values of approximately 0.90, reflecting strong overall predictive capability across dementia stages.

Despite a pronounced class imbalance within the dataset, the model demonstrated consistently high performance on the *Moderate Dementia* class, indicating reliable detection of positive dementia cases. This suggests that the incorporation of class weighting and weighted sampling contributed to improved sensitivity for underrepresented categories.

TABLE III
CLASS-WISE PERFORMANCE METRICS FOR THE BIOMARKER MODEL

Dementia Class	Precision	Recall	F1-score	Support
No Dementia	0.97	0.89	0.93	351
Dementia	0.27	0.65	0.38	23
Weighted Average	0.93	0.87	0.89	374

2) *Biomarker Model:* The biomarker-based model achieved a testing accuracy of 87.17% after 25 epochs. While overall

accuracy was lower than that of the MRI-based model, the biomarker model demonstrated strong weighted performance, achieving a weighted average F1-score of 0.89. Notably, the model exhibited high recall for the *Dementia* class (0.65), indicating moderately effective sensitivity to positive dementia cases.

However, this sensitivity was accompanied by reduced precision, reflecting an increased rate of false positives. These results highlight the biomarker model’s strength in detecting potential dementia cases, suggesting its suitability as a complementary modality in a multi-model dementia prediction framework.

3) *Speech Model*: This section presents the empirical results of the proposed multimodal dementia-detection pipeline using Speech-based models. We report fine-tuning validation results for the two pretrained encoders (BERT and wav2vec2), training behaviour and test-set performance of the fused-embedding MLP classifier, inference on a small unseen demonstration set, limitations, and a short reflection summarizing the main outcomes. Visual summaries (training curves and component comparison) are provided in Figures 1–4.

A. Encoder Fine-Tuning Performance

We fine-tuned BERT and wav2vec2 separately on the Cookie Theft dataset (551 paired samples) and monitored validation accuracy after each epoch. BERT demonstrated stronger text-level discriminative capacity compared with direct acoustic fine-tuning: the best validation accuracy observed for BERT was **70.73%** (epoch 3), whereas wav2vec2 reached a best validation accuracy of **67.07%** (epoch 5). For reproducibility, the per-epoch validation accuracies observed during training are listed below.

- **BERT fine-tuning**: Val Acc per epoch = [68.29%, 65.85%, 70.73%, 70.73%, 70.73%]; best val acc = **70.73%**.
- **wav2vec2 fine-tuning**: Val Acc per epoch = [53.66%, 52.44%, 58.54%, 60.98%, 67.07%]; best val acc = **67.07%**.

B. MLP Classifier Training and Test Performance

Fused 1536-dimensional embeddings (concatenation of 768-d BERT and 768-d wav2vec2 pooled vectors) were used to train a compact MLP classifier on an 80/20 random train/test split (seed = 42). Table IV shows sampled MLP training checkpoints and corresponding held-out test accuracies; Figure 1 visualizes test accuracy across sampled epochs and Figure 2 the corresponding training losses.

Although training loss generally decreases, test accuracy fluctuates across epochs (see Fig. 1), which suggests sensitivity to the particular train/test split and the limited sample size. The best observed test accuracy across the full training run was **91.89%**, which was saved as the final classifier checkpoint.

TABLE IV
SAMPLED MLP TRAINING CHECKPOINTS (LOSS AND HELD-OUT TEST ACCURACY).

Epoch	Loss	Test Accuracy (%)
10	0.3142	90.09
20	0.3150	90.99
30	0.1901	84.68
40	0.2408	87.39
50	0.0647	87.39
60	0.1416	89.19
70	0.1029	88.29
80	0.0325	82.88
90	0.1815	90.09
<i>Best observed test accuracy during training: 91.89%</i>		

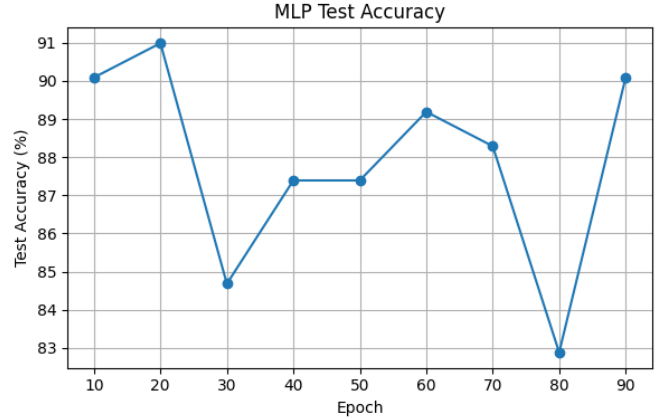


Fig. 1. MLP test accuracy at sampled epochs. The plot visualizes test performance across the training run (sampled every 10 epochs).

C. Component Comparison

Figure 3 compares the best validation/test accuracies of the individual encoders and the fused-MLP classifier. The fused approach substantially outperforms either encoder alone on the held-out test set (MLP best test accuracy = **91.89%**), while BERT and wav2vec2 alone achieved best validation accuracies of **70.73%** and **67.07%**, respectively. This suggests that the multimodal fusion captures complementary information from acoustic and lexical channels.

D. Unseen Test Inference

As a small generalization demonstration, the final pipeline was evaluated on an unseen dataset of 8 recordings (4 control recordings collected from group members and 4 dementia recordings sourced online). The model correctly labeled all eight samples (see confusion matrix in Fig. 4). Given the very small demonstration set, this perfect result is anecdotal and should not be interpreted as strong evidence of generalization.

E. Qualitative Observations

- **Fusion advantage**: Multimodal concatenation of text and audio embeddings yields substantially higher classification accuracy compared to either modality used alone on in-domain data.

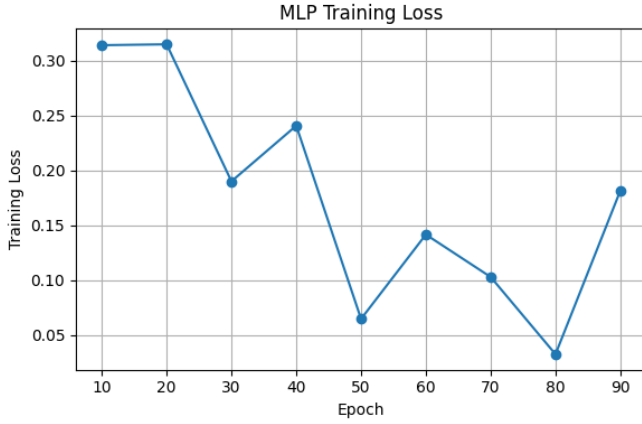


Fig. 2. MLP training loss at sampled epochs (reported losses from the training log).

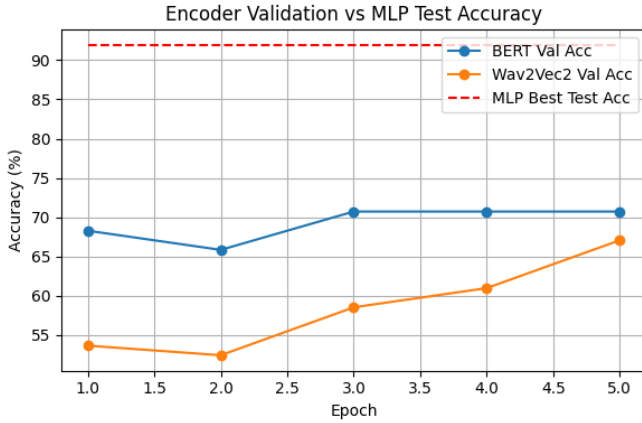


Fig. 3. Best validation accuracy for BERT and wav2vec2 (per-epoch) compared with the MLP classifier's best test accuracy.

- **Training fluctuation:** Test accuracy fluctuates across epochs, indicating sensitivity to the random split and potential overfitting / under-regularization effects on a modest dataset.
- **Encoder behavior:** wav2vec2 demonstrated a slower improvement curve compared to BERT, suggesting that acoustic fine-tuning may require more data or longer training to reach robust performance on this task.
- **Caveated perfect inference:** The unseen demonstration set was predicted perfectly; however its small and selective nature limits claims about real-world generalization.

F. Reflection

In summary, the multimodal pipeline produced strong in-domain performance: the fused embedding MLP achieved a best observed test accuracy of **91.89%** on the Cookie Theft dataset, substantially outperforming either text-only (BERT $\approx 70.7\%$) or audio-only (wav2vec2 $\approx 67.1\%$) fine-tuned encoders. While these results validate the utility of early fusion of acoustic and lexical features for this task, they should be

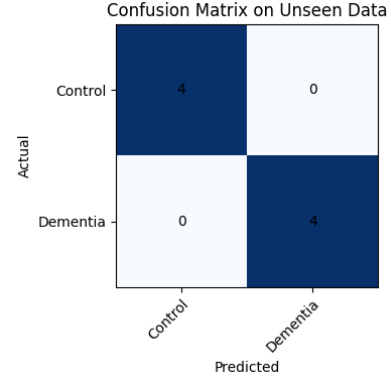


Fig. 4. Confusion matrix for the unseen demonstration set (4 control, 4 dementia). The classifier correctly labeled all 8 samples.

interpreted cautiously due to dataset size, domain narrowness, and compute/augmentation trade-offs. Future work should prioritize cross-dataset evaluation, richer and more diverse audio collections, and systematic reporting of precision/recall/F1 to better characterize clinical utility.

V. CONCLUSION & FUTURE WORK

A. Summary of Contributions

This work presents a comprehensive, multi-model approach to dementia risk prediction that integrates three complementary modalities: neuroimaging (MRI), clinical biomarkers, and speech. For each modality, we designed and evaluated a dedicated neural model tailored to the structure and constraints of the available data. The study emphasizes reproducibility, with clearly documented preprocessing, training procedures, and saved model artifacts. Collectively, this work demonstrates the feasibility and effectiveness of modular, modality-specific modeling strategies for dementia detection under limited data and compute constraints.

B. Key Findings

Across all modalities, the models demonstrated meaningful predictive capability, with performance varying according to data richness and modality characteristics. The MRI-based CNN achieved the highest single-modality accuracy, confirming the strong discriminative power of structural neuroimaging features. The biomarker-based model showed moderate overall accuracy but achieved relatively high recall for dementia cases, indicating strong sensitivity and suitability for standard screening-oriented applications. The speech-based multimodal model, which fused wav2vec2 audio embeddings and BERT text embeddings, achieved a best observed test accuracy of **91.89%**, suited for conversational analysis. These results suggest that speech contains complementary acoustic and linguistic cues relevant to dementia detection and that early fusion is an effective strategy for leveraging them.

C. Limitations

Several limitations should be considered when interpreting these results. First, all models were trained and evaluated on relatively small datasets, which increases sensitivity to random data splits and limits generalizability. Second, the datasets are domain-specific: MRI scans and Cookie Theft speech recordings may not reflect the variability present in real-world clinical settings. Third, class imbalance—particularly in the biomarker and speech datasets—was not explicitly addressed through reweighting or resampling, potentially biasing performance metrics. Finally, computational constraints imposed by the Google Colab environment necessitated training shortcuts such as micro-batching and limited fine-tuning epochs, which may have prevented full convergence of some models.

D. Future Work

Future work should prioritize improving robustness and generalizability across all modalities. This includes evaluating models on external datasets and expanding speech and biomarker data beyond the Cookie Theft task to capture more naturalistic clinical interactions. More advanced multimodal fusion strategies, such as attention-based fusion or joint end-to-end training across modalities, could further enhance performance. Addressing class imbalance through cost-sensitive learning or calibrated decision thresholds would improve clinical reliability, particularly for dementia-positive cases. Finally, future studies should report a broader set of evaluation metrics, including precision, recall, F1-score, and calibration measures, to better assess clinical utility and risk stratification performance.

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